

MATH 110

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1.1 Lecture 1 - Introduction to PDEs

What is a ODE? Recall that an ordinary differential equation (ODE) are equations for functions of a single variable, $u(t)$, $t \in \mathbb{R}$

$$\frac{du}{dt} = u(1 - u) \text{ is an ODE}$$

Here we call t the independent variable and u the state/dependent variable. In general, an ODE has the form $F(t) = (t, u, \frac{du}{dt}, \frac{d^2u}{dt^2}, \dots)$

Note that the state variable can have many components.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \in \mathbb{R}^2$$

then

$$\frac{d\mathbf{u}}{dt} = \mathbf{u} \text{ is an ODE}$$

What is a PDE? A PDE is an equation for a function with more than 1 independent variable, e.g. $u(x, y)$.

$$\begin{aligned}\text{Notation: } u_x &= \frac{\partial u}{\partial x} = \partial_x u \\ u_y &= \frac{\partial u}{\partial y} = \partial_y u \\ \text{etc.}\end{aligned}$$

Examples of PDEs

- Heat/diffusion equation for $u(x, t)$

$$u_t = k u_{xx} \quad k \text{ constant}$$

- Wave equation for $u(x, t)$

$$u_{tt} = c^2 u_{xx} \quad c \text{ constant}$$

- Laplace's equation for $u(x, y)$

$$u_{xx} + u_{yy} = 0$$

ex Consider the heat flow in a bar of length l

- $u(x, t)$ is the temperature of the bar at position x , time t
- Suppose the ends are held at temperature 0, and the initial temperature is $\phi(x)$
- The temperature evolves according to the heat equation $u_t = k u_{xx}$ where $t \geq 0, 0 \leq x \leq l, k$ constant.

We also have:

- Boundary conditions (BCs):

$$u(0, t) = u(l, t) = 0 \quad \forall t$$

- Initial conditions (ICs):

$$u(x, 0) = \phi(x)$$

Remarks

- The order of the PDE is the order of the highest derivative that occurs.
- The heat equation above is 2nd order
- A general 2nd order PDE for $u(x, t)$ is:

$$G(x, t, u, u_x, u_t, u_{xx}, u_{tt}, u_{xt}) = 0$$

- A PDE in the above form is linear if G is a linear function of u and all of its derivatives. It is nonlinear otherwise.
- A linear PDE is homogeneous if every term contains u or some derivative of u . It is nonhomogeneous/inhomogeneous otherwise.

ex Consider the linearity and homogeneity of the following equations.

1. $u_t + e^{x^2t}u_{xt} = 0$
is a linear and homogeneous PDE.
2. $u_t + 3xu_{xx} = \frac{t}{x}$
is a linear and inhomogeneous PDE because of the $\frac{t}{x}$ term.
3. $u_t + uu_x = 0$
is a nonlinear PDE because of the uu_x term.
4. $u_{tt} + u_x + \sin(u) = 0$
is nonlinear because of the $\sin(u)$ term.

Operating notation & linearity

We can write the heat equation $u_t - ku_{xx} = 0$ as $Lu = 0$ where

$$L = (\partial_t - k\partial_x^2)$$

We call L the differential operator. L acts on (smooth) functions $u(x, t)$ to produce a new function.

$$\begin{aligned} L(u) &= Lu = (\partial_t - k\partial_x^2)u \\ &= u_t - ku_{xx} \end{aligned}$$

L is a linear operator iff

$$\begin{aligned} L(u + v) &= Lu + Lv \\ \forall \text{ functions } u, v \end{aligned}$$

and

$$\begin{aligned} L(cu) &= cLu \\ \forall \text{ constants } c \in \mathbb{R} \\ \forall \text{ functions } u \end{aligned}$$

[ex] Check if the heat equation has a linear differential operator.

1.

$$\begin{aligned} L(u + v) &= (\partial_t + k\partial_x^2)(u + v) \\ &= u_t + ku_{xx} + v_t + kv_{xx} \\ &= Lu + Lv \quad \checkmark \end{aligned}$$

2.

$$\begin{aligned} L(cu) &= cu_t + cku_{xx} \\ &= cLu \text{ for constant } c \quad \checkmark \end{aligned}$$

Remarks

- If L is a linear operator then $Lu = f$ is a linear PDE

$$\underbrace{u_t + ku_{xx}}_{Lu} = \underbrace{e^{x^2t}}_{f(x,t)} \text{ is linear}$$

- If L is a linear operator then the PDE $Lu = f$ is

- homogeneous if $f = 0$
- inhomogeneous if $f \neq 0$

- A linear, homogeneous PDE $Lu = 0$ has the superposition principle

- If u_1, u_2 are solutions to the PDE then $au_1 + bu_2$ is a solution for constants a, b .

$$\begin{aligned} Lu_1 &= Lu_2 = 0 \\ \Rightarrow L(au_1 + bu_2) &= 0 \end{aligned}$$

- The distinction between linear & nonlinear PDEs is very important.

Solutions to PDEs

- A solution to a PDE is a function u in which satisfies the PDE and any BCs/ICs upon direct substitution.
- The general solution to an ODE involves arbitrary constants. The general solution to a PDE involves arbitrary functions.
- Typically, we are interested in specific solutions.

[ex] Consider the heat equation $u_t - u_{xx} = 0$.

$$\begin{aligned} u_1(x, t) &= x^2 + 2t \text{ is a soln} \\ \rightarrow \partial_t u_1 - \partial_x^2 u_1 &= 2 - 2 = 0 \\ u_2(x, t) &= e^{-t} \sin x \text{ is a soln} \\ \rightarrow \partial_t u_2 - \partial_x^2 u_2 &= -e^{-t} \sin x + e^{-t} \sin x = 0 \end{aligned}$$

[ex] Consider the equation $u_x = tx$ (1st order, linear, inhomogeneous)

$$\begin{aligned} Lu &= f \text{ with } L = \partial_x \\ f(x, t) &= tx \end{aligned}$$

We can solve this by direct integration on both sides wrt x .

$$u(x, t) = \frac{1}{2}tx^2 + A(t)$$

Where $A(t)$ is an arbitrary function of t

[ex] Consider the equation $u_{tt} - 4u = 0$ (2nd order, linear, homogeneous)

$$\begin{aligned} Lu &= f \\ \rightarrow L &= (\partial_t^2 - 4) \\ f(x, t) &= 0 \end{aligned}$$

There is no explicit x -dependence

- We can treat this as an ODE with x as a parameter.
- 'Constants' depend on x
- General solution:

$$u(x, t) = A(x)e^{-2t} + B(x)e^{2t}$$

1.2 Lecture 2 - Characteristics

Recap:

- Heat equation for $u(x, t)$

$$u_t - ku_{xx} = 0, \quad k > 0, \text{ constant}$$

- We can write the heat equation as $Lu = 0$, where $L = (\partial_t - k\partial_x^2)$ is a differential operator.
- L is linear $\leftrightarrow L(u + v) = Lu + Lv$ and $L(cu) = cLu \quad \forall$ functions u, v , constant c
- PDE: $Lu = f$ is linear if L is linear
- Linear PDE: $Lu = f$ is homogeneous if $f = 0$, inhomogeneous otherwise.

First-order linear PDE

Consider functions $u(x, y)$

$$u : \mathbb{R}^2 \rightarrow \mathbb{R}$$

Often we think of x, y as spatial variable and t as time variables. The simplest PDE we can have following this is:

$$u_x = 0$$

Which is 1st-order, linear, homogeneous. Also note that there is no explicit y -dependence. Therefore we can treat this like an ODE, and solve with direct integration of both sides.

$$u(x, y) = f(y)$$

where $f(y)$ is an arbitrary function of y .

Now consider $au_x + bu_y = 0$, which is a 1st-order, linear, homogeneous PDE with constants $a, b \in \mathbb{R}$.

$$\begin{aligned}\text{Let } V &= (a, b) \in \mathbb{R}^2 \\ \text{Recall } \nabla u &= (u_x, u_y) \\ \text{Then } V \cdot \nabla u &= (a, b) \cdot (u_x, u_y) \\ &= au_x + bu_y = 0\end{aligned}$$

$\Rightarrow$ Directional derivative in direction V is 0.

$\Rightarrow u$ is constant on lines parallel to V .

Let $V^\perp = (b, -a)$, so that $V \cdot V^\perp = 0$.

Lines parallel to V satisfy

$$\begin{aligned}V^\perp \cdot (x, y) &= \text{constant} \\ \Rightarrow bx - ay &= \text{constant}\end{aligned}$$

Then u is constant on lines parallel to V .

$$\begin{aligned}\Rightarrow u(x, y) &= f(bx - ay) \\ \text{for an arbitrary function of } f\end{aligned}$$

Check:

$$\begin{aligned}(a\partial_x + b\partial_y) f(bx - ay) \\ = af' - bf' = 0 \\ \Rightarrow f' \text{ is a derivative of } f : \mathbb{R} \rightarrow \mathbb{R}\end{aligned}$$

Characteristic Coordinates. In this example, the lines parallel to V are called characteristics. Here u is constant along the characteristics.

We can write points (x, y) in characteristic coordinates (ξ, η) . Where ξ is the x -intercept of the characteristic line, and η is the distance along that line

to the point (x, y) .

$$\text{So } x = \xi + a\eta = x(\xi, \eta)$$

$$y = b\eta = y(\xi, \eta)$$

$$\xi = x - a\eta$$

$$\eta = \frac{y}{b}$$

$$\xi = x - \frac{a}{b}y$$

$$b\xi = bx - ay$$

Transform PDE to (ξ, η) coordinates.

Chain rule:

$$\begin{aligned} \frac{\partial u}{\partial \eta} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \eta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \eta} \\ \Rightarrow u_\eta &= au_x + bu_y = 0 \end{aligned}$$

Using characteristic coordinates (ξ, η) , we've transformed to the PDE $u_\eta = 0$ for $u(\xi, \eta)$.

Solution:

$$u = f(\xi)$$

$$\text{where } \xi = \frac{1}{b}(bx - ay)$$

$$\Rightarrow u = g(bx - ay)$$

Where g is an arbitrary function.

Remarks:

- η is coordinates along characteristics
- u is constant on characteristics $\Rightarrow u_\eta = 0$
- ξ labels different characteristics. There are other choices for ξ but any choice $\tilde{\xi}$ would have $\tilde{\xi} = h(\xi)$

Method of Characteristics

Method for solving PDEs of form

$$a(x, y)u_x + b(x, y)u_y = 0 \quad \text{for } u(x, y)$$

Idea: Let $A(x, y) = (a(x, y), b(x, y))$

The directional derivative $A \cdot \nabla u = 0$

$\Rightarrow u$ is constant along curves w/ slope A .

- These are characteristic curves.
- Find coord η along chars
- Find coord ξ which labels chars
- PDE transforms to $u_\eta = 0$, general soln: $u = f(\xi)$

ex $u_x + yu_y = 0$

The directional derivative along $(1, y)$ are 0.

The chars $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = 1, \quad \frac{dy}{d\eta} = y$$

At $\eta = 0$, take $(x(0), y(0)) = (0, \xi)$

Solve:

$$\frac{dx}{d\eta} = 1, \quad x(0) = 0$$

$$\Rightarrow x = \eta$$

$$\frac{dy}{d\eta} = y, \quad y(0) = \xi$$

$$\Rightarrow y = \xi e^\eta$$

This follows that $\xi = ye^{-x}$.

$$\Rightarrow u_\eta = u_x \frac{\partial x}{\partial \eta} + u_y \frac{\partial y}{\partial \eta}$$

$$= u_x + \xi e^\eta u_y$$

$$= u_x + yu_y = 0$$

$$\Rightarrow u_\eta = 0$$

$$\Rightarrow u = f(\xi) = f(ye^{-x}) \text{ with arbitrary function } f$$

General solution:

Suppose we are given $u(0, y) = y^2$

$\Rightarrow$ Specific solution

$$u(0, y) = f(y) = y^2$$

$$\Rightarrow u(x, y) = (ye^{-x})^2$$

Alternatively, from parametric equations,

$$\frac{dx}{d\eta} = 1 \quad \frac{dy}{d\eta} = y$$

$$\Rightarrow \frac{dy}{dx} = y$$

$$\Rightarrow y = Ce^x$$

$\Rightarrow u$ is constant along chars.

$$\Rightarrow u(x, Ce^x) = u(0, C)$$

$$= u(0, ye^{-x})$$

$$= f(e^{-x}y) \text{ for arbitrary } f$$

$$\boxed{\text{ex}} \quad u_x + 2xy^2u_y = 0$$

Characteristic curves: $(x(\eta), y(\eta))$

$$\frac{dx}{d\eta} = 1 \quad \frac{dy}{d\eta} = 2xy^2$$

$$\frac{dy}{dx} = 2xy^2$$

$$\int \frac{1}{y^2} dy = \int 2x dx$$

$$-\frac{1}{y} = x^2 - c$$

$$y = \frac{1}{c - x^2}$$

u is constant along chars.

$\Rightarrow u(x, y) = f(c) = f(x^2 + \frac{1}{y})$ is the general soln.

Advection, transport

Consider $u(x, t)$ describing quantity u at position x , time t .

$$u_t + cu_x = 0 \quad (c > 0)$$

Chars: $(t(\eta), x(\eta))$ satisfy

$$\begin{aligned} \frac{dt}{d\eta} &= 1 & \frac{dx}{d\eta} &= c \\ \Rightarrow \frac{dx}{dt} &= c \\ \Rightarrow x &= ct + \xi \text{ via integration} \end{aligned}$$

General solution:

$$u = f(\xi) = f(x - ct)$$

1.3 Lecture 3 - Conservation Laws

Recap:

- Solve PDEs of the form $a(x, y)u_x + b(x, y)u_y = 0$
- The char curves $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = a(x, y) \quad \frac{dy}{d\eta} = b(x, y)$$

- u is constant along chars.

ex Given $\frac{1}{x^2}u_x + u_y = 0$

$$\frac{dx}{d\eta} = \frac{1}{x^2} \quad \frac{dy}{d\eta} = 1$$

Method 1: divide to get parametric eq.

$$\begin{aligned} \frac{dy}{dx} &= x^2 \\ y &= \frac{1}{3}x^3 + \xi \end{aligned}$$

where constant ξ labels the char.

The general solution is $u = f(\xi) = f(y - \frac{1}{3}x^3)$ for an arbitrary function f . This can be confirmed by plugging u back into the PDE.

Method 2: solve parametric equations.

$$\begin{aligned}\frac{dx}{d\eta} &= \frac{1}{x^2} & \frac{dy}{d\eta} &= 1 \\ \Rightarrow \int x^2 dx &= d\eta & y &= \eta + B \\ \Rightarrow \frac{1}{3}x^3 &= \eta + A & y &= \eta + B\end{aligned}$$

We can absorb the two constants into one ξ

$$\begin{aligned}\Rightarrow \frac{1}{3}x^3 &= \eta + \xi & y &= \eta \\ \Rightarrow y &= \frac{1}{3}x^3 - \xi\end{aligned}$$

We should check that the chars cover the whole plane, i.e. no chars have been 'missed', via sketching or finding

$$\begin{aligned}\eta &= \eta(x, y) \\ \xi &= \xi(x, y)\end{aligned}$$

Coord transformation is also useful

$$\begin{aligned}\frac{1}{3}x^3 &= \eta + \xi \\ y &= \eta\end{aligned} \Rightarrow \begin{aligned}x &= (3\eta + 3\xi)^{\frac{1}{3}} \\ y &= \eta\end{aligned}$$

This transform our PDE

$$\begin{aligned}\frac{1}{x^2}u_x + u_y &= 0 \Leftrightarrow u_\eta = 0 \\ \text{Gen soln: } u &= f(\xi) \\ &= f\left(\frac{1}{3}x^3 - y\right)\end{aligned}$$

with arbitrary function f

ex Advection

Consider function $u(x, t)$ describing a quantity u at position x and time t

$$u_t + cu_x = 0 \quad (\text{Advection})$$

with constant $c > 0$

Chars $(t(\eta), x(\eta))$ satisfy

$$\frac{dt}{d\eta} = 1 \quad \frac{dx}{d\eta} = c$$

Solve

$$t = \eta + A$$

$$x = c\eta + B$$

Absorb into one constant using $A = 0, B = \xi$

$$\Rightarrow t = \eta$$

$$x = c\eta + \xi$$

$$\Rightarrow x = ct + \xi$$

By inverting, $\eta = t, \xi = x - ct$.

Now we have an transformed PDE

$$u_\eta = u_t + cu_x = 0$$

General solution is

$$\begin{aligned} u(x, t) &= f(\xi) \\ &= f(x - ct) \end{aligned}$$

for an arbitrary f .

Suppose we had initial condition $u(x, 0) = \phi(x)$, then $u(x, 0) = f(x) = \phi(x) \Rightarrow$ solution is $u(x, t) = \phi(x - ct)$.

Remarks:

We can solve more complicated PDEs using chars, for example,

$$u_t + cu_x = f(u)$$

Transform to coords (ξ, η) with $t = \eta$, $x = c\eta + \xi$

$$u\eta = f(u)$$

which can be solved as an ODE

Conservation Laws. (Logan 1.2)

- Where do PDEs come from?
- For a conserved quantity:

$$\begin{array}{ccc} \text{rate of change} & \implies & \text{net flow into domain} \\ \text{in domain} & & + \text{creation in domain} \\ \text{e.g. Animal population in a fixed region} & & \end{array}$$

Quantitatively

- Let $u(x, t)$ be density of some quantity in 1d domain.
- The amount of quantity in section dx is $u(x, t)Adx$.
- Let $\phi(x, t)$ be the flux, which is the amount of quantity crossing section x at time t .
- $A\phi(x, t)$ is amount of quantity passing x at time t
- Let $f(x, t)$ be the rate at which quantity is created.
- $f(x, t)Adx$ is amount of quantity created in width dx per unit time.

Remarks:

- $\phi(x, t) > 0 \implies$ flux is L to R
- $\phi(x, t) < 0 \implies$ flux is R to L
- $f(x, t) > 0$ is source at x (creation)
- $f(x, t) < 0$ is sink at x (destroyed)

Balance:

change in
quantity in = flux + creation
section $[a, b]$

$$\begin{aligned}\Rightarrow \frac{d}{dt} \int_a^b u(x, t) A dx &= A\phi(a, t) - A\phi(b, t) + \int_a^b f(x, t) A dx \\ \Rightarrow \int_a^b u_t A dx &= \int_a^b \phi_x A dx + \int_a^b f A dx \\ \Rightarrow \int_a^b (u_t - \phi_x - f) dx &= 0\end{aligned}$$

This is true for any domain $[a, b]$

$u_t = \phi + f$ where
change in
quantity = net flux + net creation

Remarks:

Can get a PDE for u by

- Specifying $f(x, t)$
- Choosing a relationship between ϕ and u, u_x , etc. This is called constitutive law.
- E.g. $\phi = cu \rightarrow$ advection.

Diffusion:

- Conservation law with no sources

$$u_t + \phi_x = 0$$

- Suppose $u(x, t)$ is density of a gas of particles. Then
 - Particles move from high u to low u .

- Flux increase with density gradient.
- Fick's law:

$$\phi(x, t) = -Du_x$$

where $D > 0$ is diffusion constant.

- This gives the diffusion equation

$$u_t = Du_{xx}$$

which is the same as the heat equation.

Remark:

- Can combine diffusion with other effects.
e.g. Advection-diffusion eq.

$$\text{flux: } \phi = cu - Du_x$$

Balance:

$$u_t + \phi_x = 0$$

$$u_t + cu_x = Du_{xx}$$